STANDARD OPERATING PROCEDURE
(SOP)
WEMBIYAGODA MINI HYDRO POWER PLANT
(WMB MHPP)

Contents

Standard Operating Procedure

Function Responsibilities

1. Purpose

This document will guide function wise responsible persons to carry out all necessary work **related to Mini hydro Power plant under Vidul group.** This will explain processes/Functions and involvements of direct workers, Supervisors, Area Engineers, Operation Manager and all Service

විදුල් සමූහය යටතේ මිනි ජල විදුලි බලාගාරයට අදාළ සියලු කටයුතු සිදු කිරීමට මෙම ලේඛනය ක්‍රියාකාරී බුද්ධිමත් වගකිවයුතු පුද්ගලයින්ට මග පෙන්වනු ඇත. සෘජු සේවකයින්, අධීක්ෂකවරුන්, ප්‍රාදේශීය ඉංජිනේරුවන්, මෙහෙයුම් කළමනාකරු සහ සියලුම සේවා දෙපාර්තමේන්තු වල ක්‍රියාවලි / කාර්යයන් සහ සහභාගීත්වය මෙයින් පැහැදිලි කෙරේ.

2. Scope

All helpers, operators, Supervisors or relevant engineers are supposed to follow the procedures and schedules. No person should bypass the responsibilities.

සියළුම සහායකයින්, ක්‍රියාකරුවන්, අධීක්ෂකවරුන් සහ අදාළ ඉංජිනේරුවන් විසින් ක්‍රියා පටිපාටි සහ උපලේඛන අනුගමනය කළ යුතුය. කිසිම පුද්ගලයෙකුට වගකීම් මඟ හැරිය නොහැක.

3. Prerequisites

All responsibilities have set relevant to each plant and all formats to be used as given by the Head office's Operation department only.

සියළුම වගකීම් එක් එක් බලාගාරයට අදාළ වන අතර, ප්‍රධාන කාර්යාලයේ මෙහෙයුම් දෙපාර්තමේන්තුව විසින් ලබා දී ඇති ආකෘති පමණක් භාවිතා කළ යුතුය.

4. Responsibilities - වගකීම්

Engineering department directly responsible for the operation procedures and all helpers, operators and supervisor will be responsible with the operation.

මෙහෙයුම් ක්‍රියා පටිපාටි සඳහා ඉංජිනේරු දෙපාර්තමේන්තුව සෘජුවම වගකිව යුතු අතර මෙහෙයුමට සියලු සහායකයින්, ක්‍රියාකරුවන් සහ අධීක්ෂක වගකිව යුතුය.

5. Functions\Systems & Relevant persons with Designations කාර්යයන්\ පද්ධති

No.	Functions\Systems	Frequency	Level 1- Name	Level 2 - Name	Level 3 – Name\s
			Area Engineer		
01	Inventory System – Updates ඉන්වෙන්ටරි පද්ධතිය - යාවත්කාලීන කිරීම්	Daily දිනපතා			
02	HRIS Updates- Attendance \Leave HRIS යාවත්කාලීන කිරීම්- පැමිණීම \ නිවාඩු	Daily දිනපතා			
03	Monthly meeting – 5S මාසික රැස්වීම - එස් 5	Monthly මාසික			
04	5S Audit එස් 5 විගණනය	Monthly මාසික			
05	Patrol Team මුර සංචාර කණ්ඩායම	Weekly සතිපතා			
06	Red\Yellow tagging රතු \ කහ ටැගින් කිරීම	Monthly මාසික			
07	Safety Audit (Safety Officers) ආරක්ෂිත විගණනය (ආරක්ෂක නිලධාරීන්)	Monthly මාසික			
08	Welfare management සුභසාධන කළමනාකරණය	Monthly මාසික			
09	Fire Extinguishers ගිනි නිවන උපකරණ	Yearly වාර්ෂික			
10	Crane දොඹකරය	Weekly සතිපතා			
11	Notices & Notice Board දැන්වීම්	Weekly සතිපතා			
12	Reforestation වන වගාව	Monthly මාසික			
13	Plastic free Implementation ප්ලාස්ටික් වලින් තොර පරිසරයකට ගමන් කිරීම	Monthly මාසික			
14	Waste Management- අපද්‍රව්‍ය කළමනාකරණය	Monthly මාසික			

15	Meeting Minute Update, රැස්වීම් තොරතුරු යාවත්කාලීන කිරීම	Weekly සතිපතා			

6. *References - යොමුව*

Refers to the guidelines given by Operation Department - HOF
මෙහෙයුම් දෙපාර්තමේන්තුව - HOF විසින් ලබා දී ඇති මාර්ගෝපදේශ වෙත යොමු වේ

7. *Note - සටහන*

All explanation are attached in native (Sinhala\Tamil) languages for understanding purpose.
සියලු පැහැදිලි කිරීම් අවබෝධ කර ගැනීම සඳහා ස්වදේශීය (සිංහල \ දෙමළ) භාෂාවලින් අමුණා ඇත.

Standard Operating Procedure

Location Responsibilities

1. Purpose

This document will guide on persons required to carry work **responsibilities location-wise.** This will explain involvements of direct workers, Supervisors, Area Engineers, Operation Manager and all Service.

ස්ථානය අනුව වැඩ කිරීමේ වගකීම් ඉටු කිරීමට අවශ්ය පුද්ගලයින්ට මෙම ලේඛනය මග පෙන්වනු ඇත. සෘජු සේවකයින්, අධීක්ෂකවරුන්, ප්රාදේශීය ඉංජිනේරුවන්, මෙහෙයුම් කළමණාකරු සහ සියලුම සේවාවන් සම්බන්ධ කර ගැනීම මෙයින් පැහැදිලි කෙරේ.

2. Scope

All helpers, operators, Supervisors and relevant engineers are supposed to follow the procedures. No person should bypass the responsibilities.

සියලුම සහායකයින්, ක්රියාකරුවන්, අධීක්ෂකවරුන් සහ අදාළ ඉංජිනේරුවන් විසින් ක්රියා පටිපාටීන් අනුගමනය කළ යුතුය. කිසිම පුද්ගලයෙකු වගකීම් මඟ හැරිය යුතු නැත.

3. Prerequisites

All responsibilities have set relevant to each plant and all formats to be used as given by the Head office's Operation department only.

සියළුම වගකීම් එක් එක් බලාගාරයට අදාළ වන අතර, ප්රධාන කාර්යාලයේ මෙහෙයුම් දෙපාර්තමේන්තුව විසින් ලබා දී ඇති ආකෘති පමණක් භාවිතා කළ යුතුය.

4. Responsibilities

Engineering department directly responsible for the operation procedures and all helpers, operators and supervisor will be responsible with the operation.

මෙහෙයුම් ක්රියා පටිපාටි සඳහා ඉංජිනේරු දෙපාර්තමේන්තුව සෘජුවම වගකිව යුතු අතර මෙහෙයුමට සියලු සහායකයින්, ක්රියාකරුවන් සහ අධීක්ෂක වගකිව යුතුය.

5. Locations & Relevant persons with Designations
තනතුරු සහ තනතුරු සමඟ අදාළ පුද්ගලයින්

No.	Location	Level 1-Name Area Engineer	Level 2 - Name Supervisor	Level 3 – Name\s
01	Machine Floor යන්ත්ර මහල			
02	Office කාර්යාලය			
03	Control Room පාලන කාමරය			
04	Meeting Room රැස්වීම් කාමරය			
05	Stores ගබඩාව			
06	Bath Room\Toilet නාන කාමරය \ වැසිකිළිය			
07	Switch Yard වහරු අංගනය			
08	Garden වත්ත			
09	Bike & Car park බයික් සහ කාර් පාක්			

10	Quarters නිල නිවස			
11	Dining Room කෑම කාමරය			
12	Kitchen මුළුතැන්ගෙය ෆෝබේ ටැංකිය			
13	Forby Tank ෆෝබි ටැංකිය			
14	Channel Area ඇල මාර්ගය			
15	Intake\Weir වේල්ල			

6. References - යොමුව

Refers to the guidelines given by Operation Department - HOF
මෙහෙයුම් දෙපාර්තමේන්තුව - HOF විසින් ලබා දී ඇති මාර්ගෝපදේශ වෙත යොමු වේ

7. Note - සටහන

All explanation are attached in native (Sinhala\Tamil) languages for understanding purpose.
සියලු පැහැදිලි කිරීම් අවබෝධ කර ගැනීම සඳහා ස්වදේශීය (සිංහල \ දෙමළ) භාෂාවලින් අමුණා ඇත.

1.Standard Operation Procedure for Lock out Tagout System

1.1. Purpose

This document will guide on how to do the operation and maintenance work according to the lockout tagout system in a safe way.

1.2. Scope

Should be followed lockout tagout system while doing operation and maintenance work.

1.3. Prerequisites

Should be identified equipments with hazardous energy sources of the site.

1.4. Responsibilities

The supervisor and site staff should be responsible.

1.5. Procedure

1. Prepare for shutdown.
2. Notify all affected employees of the activities and equipment involved.
3. Shutdown the equipment.
4. Isolate the equipment from hazardous energy sources.
5. Dissipate residual energy.
6. Apply applicable lockout tagout device.
7. Verify that the equipment is properly isolated.

2.Standard Operating Startup, Stop and Emergency Stop Procedure

2.1. Purpose

This document will guide on, how to Start and Stop a machine manually and on an emergency in a methodical way.

2.2. Scope

Only qualified people are supposed to operate.

2.3. Prerequisites

Water level at forebay tank, oil levels and oil pressure on relevant units and brake pads status to be checked.

2.4. Responsibilities

The supervisor/operators should be responsible.

2.5. Procedure.

2.5.1 Startup Procedure.

1. Check the forebay water level.
2. Close the DDLO set.
3. Check the bearing oil level (It should be half of indicator glass) and oil pressure unit oil level (It should be up to filter level).
4. Switch on the Governor pressure pump and check the pressure is within limit. (9MPa-12MPa)
5. Open the Bypass valve and air release valve. Close the valve after the trapped air is released.
6. Ensure the pressure of both sides of the MIV valve.

7. Open the MIV valve.
8. Open and check the bearing cooling water supply.
9. Release the brake and check brake pads.
10. Start the machine.
11. Ensure the flow sensor and temperature sensors are working properly.
12. Turn the selector switch to grid, in main panel.
13. AVR will be switched on and the synchronization will be automatically.
14. Increase the load with respect to the forebay water level communication or heads level sensor.

2. 5.2. Stop Procedure.

1. Decrease the load to 100kW.
2. Trip the SYN-breaker.
3. Close MIV and Bypass valve.
4. Apply brakes to the machine when machine rpm is reduced below 100rpm.
5. Stop the Governor pressure pump.

2.5.3. Emergency Stop Procedure.

1. Push the Emergency Button.
2. Close MIV and Bypass valve.
3. Apply brakes to the machine when machine rpm is reduced below 100rpm.
4. Stop the Governor pressure pump.
5. Open the HVCB. (If necessary.)
6. Open the relevant DDLO set of the transformer. (If necessary.)

3. Standard Start-up operation after the short-term suspension of operation (within 24 hours)

3.1. Purpose

This document will guide you on how to do the start-up operation after short term suspension (within 24hours) of a machine in a methodical way.

3.2. Scope

Only qualified people are supposed to operate.

3.3. Prerequisites

Water level at forebay tank, oil levels and oil pressure on relevant units and brake pads status to be checked.

3.4. Responsibilities

Supervisor/operators should be responsible.

3.5. Startup Procedure.

1. Check the function of ACB (Synchronous breaker UVT & closing coil).
2. Check the grid availability.
3. Check the enough water to run the machine.
4. Visually check the forebay water level to match with control panel or Scada water level.
5. Check any water leakages.
6. Check any oil leakages and oil levels.
7. Check the brake system status.

10. Check the Governor and bearing oil levels are in standard mentioned level (Bearing oil level should be half of indicator glass and governor oil level should be up to filter level.)
11. Check the cooling water flow
12. Switch on the governor pump
13. Check the governor's pressure (It should be in 9-12MPa range).
14. Open the bypass valve.
15. Open the Main inlet valve (MIV).
16. Start the machine and check for any abnormalities.
17. Build up the voltage and run without synchronizing for half an hour.
18. Synchronous machine with grid and load 30% of the rated output power for one hour.
19. Check any abnormalities (Current, voltage, power factor, etc.).
20. Increase the power to 100kW and run for half an hour.
21. Mean time check the temperature values of generator windings, generator side bearings and turbine side bearings etc.
22. Gradually increase load while winding temperature is increased by 10 centigrade rather than ambient temperature and run the machine with above mention time and power.
23. Mean time check is there enough water level to load the machine with optimum machine loading combination.

4. Standard operation procedure for power transformer testing and cleaning work.

4.1. Purpose

This document will guide you on how to do the power transformer testing and cleaning work in a methodical way.

4.2. Scope

Electrical Engineer and the relevant supervisor, operators should be Involved.

4.3. Prerequisites

The area engineer must be notified first and approved by the Chief Electrical Engineer.Power supply should be disconnected for relevant transformer. Transformer terminals should be earthed using discharge rod. People should wear PPE. A tool assembly should be set up before work begins.

4.4. Responsibilities

Under the supervision of the Electrical Engineer, work should be done.

4.5. Procedure

1. Be careful not to do unsafe and dangerous work.
2. Follow the Lockout tagout procedure
3. Use the backup power to powerhouse usages.

4. Carefully remove the power cables (LV & HV).
5. Carrying out the IR test and values should be confirmed by Chief Electrical Engineer.

10. Check the oil level, any oil leak and insulators properly, is there any abnormalities, inform it to Chief Electrical Engineer as soon.
11. After finished the cleaning process and testing carefully fix the power cables.
12. Inspection should be done by the Electrical Engineer and get the confirmation to fix the DDLO.
13. Take pictures of the process and send them to the relevant authorities.

5.Standard operation procedure of HVCB testing and cleaning work.

5.1. Purpose

This document will guide you on how to do the HVCB testing and cleaning work in a methodical way.

5.2. Scope

Electrical Engineer and the relevant supervisor, operators should be Involved.

5.3. Prerequisites

The area engineer must be notified first and approved by the chief electrical engineer. A tool assembly should be set up before work begins. Power must be disconnected with removing DDLO and send it to the DDLO cabin. Have the necessary equipment ready before starting the work. Properly earth all terminals before starting cleaning work.

5.4. Responsibilities

Under the supervision of the Electrical Engineer, work should be done.

5.5. Procedure

1. Be careful not to do unsafe and dangerous work.
2. Follow the Lock out tagout procedure.
3. Carefully check the tightness of all control cables.

4. Must be check the heating equipment condition and spring charging equipment and process.
5. Use the backup power to check the function and powerhouse usages.
5. Check the on-off function two three times with local and remote.
6. Carrying out the IR test and values should be confirmed by Chief Electrical Engineer
7. Check the insulators properly and is there any abnormalities, inform it to Chief Electrical Engineer as soon.
8. After finished the cleaning process and testing, inspection should be done by the Electrical Engineer and get the confirmation to fix the DDLO.
9. Take pictures of the process and send them to the relevant authorities.

6.Standard operation procedure of CT PT Cleaning

6.1. Purpose

This document will guide you on how to do the CT PT cleaning work in a methodical way.

6.2. Scope

Electrical Engineer and the relevant supervisor, operators should be Involved.

6.3. Prerequisites

The area engineer must be notified first and approved by the chief electrical engineer. A tool assembly should be set up before work begins. Power must be disconnected with removing DDLO and keep it in DDLO cabin. Have the necessary equipment ready before starting the work. Properly earth all terminals before starting cleaning work.

6.4. Responsibilities

Under the supervision of the Electrical Engineer, work should be done.

6.5. Procedure

1. Use the backup power to powerhouse usages.
2. Be careful not to do unsafe and dangerous work.
3. Always use ladders and safety belts when climbing.
4. There must be someone to hold it when using the ladder.
5. Check the oil level and insulators properly and is there any

abnormalities, inform it to area engineer as soon.

6. After finished the cleaning process, inspection should be done by the supervisor and get the confirmation from Electrical engineer and fix the DDLO.
7. Take pictures of the process and send them to the relevant authorities.

7.Standard operation procedure of Intake gate

7.1. Purpose

This document will guide you on how to do the Intake gate operation in a methodical way.

7.2. Scope

Relevant intake gate operator should coordinate with supervisor to operate the gate.

7.3. Prerequisites

Well trained people should be involved, and PPE should be worn.

7.4. Responsibilities

The supervisor should be responsible.

7.5. Procedure

7.5.1. Opening the Gate:

1. Gradually open the gate 100mm by 100mm using the handle, monitoring the gate's movement and water flow rate.
2. Communicate with the Forebay to confirm the gate is opening smoothly and maintain the water level without spill and maintain that level at the machines are running full load condition.
3. Once the gate reaches the desired position, lock it in place if necessary.

7.5.2. Closing the Gate:

1. Communicate with the Forebay to begin the closing procedure.

2. Should ensure that the gate is fully closed.
3. The gate should be closed properly without applying additional force.
4. Lock the gate in the closed position if required.

7.5.3. Post-Operation Procedure

1. Perform a visual inspection of the gate and surrounding area.
2. Record the operation in the logbook, noting any issues or irregularities.
3. Report any maintenance needs to the appropriate personnel.
4. Restore any equipment used to its proper storage location.

7.5.4. Emergency Procedures

1. In case of gate malfunction, immediately engage the emergency stop mechanism.
2. Notify the control room and maintenance team of the issue.
3. Follow the plant's emergency response plan.

7.5.5. Documentation and Record Keeping

1. Maintain a logbook of all gate operations, including date, time, operator name, and any issues encountered.
2. Keep records of maintenance activities, inspections, and repairs.

8.Standard operation procedure of forebay sand removing

8.1. Purpose

This document will guide you on how to do the forebay sand removing in a methodical way.

8.2. Scope

Supervisor should take necessary action for the operation.

8.3. Prerequisites

Check whether that sand level exceeded predetermined value. (If exceed 1m). A tool assembly should be set up before work begins. Have the necessary equipment ready before closing the gate.

8.4. Responsibilities

The supervisor should be responsible.

8.5. Procedure

1. The intake door should be closed, and the handle removed for safety.
2. Perform dewatering through the machine to save time.
3. The intake care person should be responsible for gate operation with the supervisor.
4. The sand removal gates should be open at the lowest water level in the channel.
5. Sand removal should be done with minimum time.

6. Before refilling the channel, should be checked the channel path.
7. The gate should be opened by 100mm by 100mm slowly to refill the channel by intake care person coordination with the supervisor.

9.Standard operation procedure of Sand gate operation

9.1. Purpose

This document will guide you on how to do the sand gate operation in a methodical way.

9.2. Scope

The supervisor should take the necessary action for this operation.

9.3. Prerequisites

Check the waterflow at the intake.

9.4. Responsibilities

Under the supervision of the supervisor, work should be done.

9.5. Procedure

1. If water overflows at the intake when the plant is running at full capacity, Open the sand gate from 2 inches. When the water level increases, open 2 inches at a time.
2. If the water does not overflow at the intake when the plant is running at full capacity, the sand gate should be closed.

10.Standard operation procedure of MIV operation

10.1. Purpose

This document will guide you on how to do the MIV operation in a methodical way.

10.2. Scope

Only supervisors, operators, knowledgably people are supposed for the operation.

10.3. Prerequisites

Check whether the bypass valve is closed, and turbine side pressure and penstock side pressure are balanced.

10.4. Responsibilities

The supervisor should be responsible.

10.5. Procedure

10.5.1 MIV Opening

1. First open the bypass valve.
2. Open the automatic / Manual MIV valve.
3. Check that the MIV valve is fully open. (90 degrees)

10.5.2.MIV Closing

1. Auto/Manual Close the MIV.
2. Check that MIV closes properly to 90 degrees.
3. Close the bypass valve.
4. Ensure that MIV is properly closed by opening the drain valve.

10.6. Notes

Please be careful to open only the machines with MIV and bypass operation and keep the others completely closed.

11.Standard operation procedure of cooling operation

11.1. Purpose

This document will guide you on how to do the cooling operation in a methodical way.

11.2. Scope

Supervisor/operator and relevant people are supposed to operate.

11.3. Prerequisites

Check whether there are any blockages in the cooling line.

11.4. Responsibilities

The supervisor should be responsible.

11.5. Procedure

11.5.1. Before starting machine

1. Only the running machines open the cooling valves and keep the others closed.
2. Ensure that water is draining in the water outlet of relevant cooling line.
3. When opening the cooling, first open the cooling valve on the penstock side and backwash the filter before starting the machine.
4. After completion of the backwashing, the cooling valve on the side of the machine should be opened.

12. Standard operation procedure of governor operation.

12.1. Purpose

This document will guide you on how to do the governor operation in a methodical way.

12.2. Scope

Only supervisors or operators are supposed to operate.

12.3. Prerequisites

Check condition of the hydraulic hose and pressure meter in visible.

12.4. Responsibilities

The supervisor should be responsible.

12.5. Procedure

1. Check that the governor's oil level is at the required level within the limit.
2. The accumulator pressure release valve should be kept closed.
3. Check that the system pressure valve is open.
4. Check for oil leakage from valves/hoses/actuator/anywhere.
5. Start the governor pump motor through the panel. (Check if the pump motor is working properly)
6. Check if the pressure is increasing on the governor pressure gauge. (Check if the maximum operating pressure is reached and the motor should automatically shut off after reaching the maximum operating pressure.) Governor pressure range should be in

9-12MPa.

7. Check the brake solenoid release (can be detected by inspecting the brake pads on the flywheel side.)
8. Do not release accumulator pressure except in an emergency during machine stopped periods.

When CEB Power Failure

The system pressure valve of the governor must be closed to maintain the oil pressure in the system. Please close the system pressure valve after 10 minutes by confirmation with CEB power.

12.6. Notes

Supervisor/Operator monitor daily oil level and pressure of governors. (In case of any abnormal oil loss of the governor please contact the Area Engineer/Mechanical Engineer concerned.)

13.Standard operation procedure of crane operation.

13.1. Purpose

This document will guide you on how to do the crane operation in a methodical way.

13.2. Scope

Only supervisors or operators are supposed to do the operation.

13.3. Prerequisites

When operating the crane, the people who are involved in the operation should wear the PPE and well-trained people only can operate the crane.

13.4. Responsibilities

The supervisor should be responsible.

13.5. Procedure

1. It is necessary to check the movements of the crane in all directions by remote control of the crane before lifting the load.
2. When your job is done, park the crane in the loading bay.

13.6. Notes.

1.Do not stand under the crane when lifting weights.
2.Try to carry the load by crane over the free area.
3.The installed crane can be only lifted loads vertically and do not lift side loads.

4. Use relevant lifting belts, D-shackles and should properly tight them and place the lifting belts so that the weight is balanced.

14.Standard operation procedure of trash machine operation.

14.1. Purpose

This document will guide you on how to do the trash machine operation in a methodical way.

14.2. Scope

Forebay care person is supposed to do the operation.

14.3. Prerequisites

Check whether the governor's pressure has reached to a specific level, check the limits switches work properly. A well-trained person should operate and need to wear PPE.

14.4. Responsibilities

The supervisor should be responsible.

14.5. Procedure

1. Use the remote controller and electrically machine can be operated and be aware of the direction on the arm which is moving.
2. Put the trash arm to the forebay channel and collect the debris and take the arm slowly along the trash rack.

14.6. Notes

1. When the trash machine is moving, work should not be done under the trash arm.
2. Do not work under the lower step area when moving the trash arm.

3.Your attention should be directed to the machine itself until it stops properly.

4. While the machine is running, do not stand on the lower floor to remove the trash.

5. Please be aware not to lift heavy load particles with trash arm.

6. Trash arm should be placed on the upper step of the trash rack at the time that is not used.

15.Standard operation procedure for water leakage in penstock.

15.1. Purpose

This document will guide you on how to take necessary actions in case of water leakage in penstock in a methodical way.

15.2. Scope

In this case, the person in charge of the plant should take the necessary action. Repair work should be done under the supervision of a supervisor and Mechanical engineer.

15.3. Prerequisites

Should be informed to the Mechanical Engineer. Should arrange the necessary tools for the work.

15.4. Responsibilities

Under the supervision of the Mechanical Engineer, work should be done.

15.5. Procedure

1.Do not stop the machines if they are running.

2.Take action to close the main gate as soon as possible.

3.If the water leakage is large, put all machines in running state.

4.Remove water through turbines and cooling lines.

5.Securely open the sand gates and drain the water.

16.Standard operation procedure for abnormal water leakage in turbine

16.1. Purpose

This document will guide you on how to take necessary actions in case of abnormal water leakage in turbine in a methodical way.

16.2. Scope

In this case, the person in charge of the plant should take the necessary action.

16.3. Prerequisites

Should be informed to the Mechanical Engineer. Arranged the required tools and materials for the task.

16.4. Responsibilities

Under the supervision of the Mechanical Engineer or supervisor, work should be done.

16.5. Procedure

1.Check the wicket gates and shear pins condition.

2.Take steps to electrically close the main inlet valve as soon as possible.

3.It is necessary to use lockout tagout system.

4.If the water leak is not controlled, try to close the valve manually.

5.Open the drain valve to control the machine speed.

6.Do not brake when driving at high speed. Use a wooden bar to fly wheel and try to control the speed.

7.Be careful not to do unsafe and dangerous work.

8.If an uncontrollable volume of water is coming out, empty the channel and check.

9.Inform the Engineers in charge immediately.

17.Standard operation procedure for water leakage to powerhouse due to flood

17.1. Purpose

This document will guide you on how to take necessary actions in case of water leakage to powerhouse due to flood in a methodical way.

17.2. Scope

In this case, the person in charge of the plant should take the necessary action.

17.3. Prerequisites

Should be aware of the daily weather forecast.

17.4. Responsibilities

The Engineering Department is directly responsible for this scenario and under the supervision of the Area Engineer, engineering team and supervisor, work should be done.

17.5. Procedure

1. Stop the machines if they are running and the intake gate should be closed immediately.
2. Take action to close the main inlet valve as soon as possible.
3. If there are any water inlets which take the chance to enter the water inside to the powerhouse should be closed.

3. Should inform the relevant engineer or manager.
4. Should remove the specific critical items as per the instructions of relevant engineer.
5. Drive site and personal vehicles to safe place.
6. Make ready stand by generator to use for power up (If necessary).
7. If a water leak can be controlled by pump, need to run them if it is required further.

18. Standard operation procedure in electrical fire

18.1. Purpose

This document will guide you on how to take necessary actions in case of electrical fire in a methodical way.

18.2. Scope

In this case, the person in charge of the plant should take the necessary action.

18.3. Prerequisites

There should be well-trained people who are aware of firefighting.

18.4. Responsibilities

The Engineering Department is directly responsible for this scenario and under the supervision of the Area Engineer, engineering team and supervisor, work should be done.

18.5. Procedure

1.Disconnect power supply relevant items immediately.

2.If it is coming from a machine, stop the machine.

3.If it continues to burn, use the suitable fire extinguisher .The way of operating fire extinguishers and the fire extinguisher types are mentioned in the location which fire extinguishers are placed.

19. Standard operation procedure for channel inspection

19.1. Purpose

This document will guide you on how to inspect the channel of the power plant in a methodical way.

19.2. Scope

In this case, the person in charge of the plant should take the necessary action.

19.3. Prerequisites

The people who attain the inspection should wear PPE.Prepare the necessary tools and a checklist for the inspections.

19.4. Responsibilities

Under the supervision of the supervisor, inspection should be done.

19.5. Procedure

19.5.1. General Observations

1. Look for visible signs of structural damage, cracks or erosion.
2. Check that there are any abnormal water leakages through the channel path.
3. Check that there are any obstacles at the channel area. (Rock/tree)
4. Check the condition of the water drainage pipes.

19.5.2. Blockages Observations

1. Inspect for debris, silt or foreign objects in the channel.
2. If safe to do so, clear minor blockages immediately using

appropriate tools.

19.5.3. Water flow observations

1. Observe the uniformity and speed of water flow.
2. Note any areas with turbulence or significantly reduced flow.

19.5.4. Structural Components observations

Intake Structure

1. Check screens and gates for damage or clogging.

Channel Wall

2. Look for leaks, cracks, or weakening of materials.

19.5.5. Weir Side Inspections

1. Check the log barrier is in good condition and check any damages of trash mesh.
2. If sand level is increased inside the weir need to take necessary action to remove them.

19.6 Notes

If any abnormal condition is observed in above mentioned aspects, should be informed to the relevant Engineer or supervisor.

General Note

Please Arrange the shutdown permit while doing testing and maintenance works which are directly affected for the power generation of the power plant.

20. Standard Operating Procedure – Authority level for Plant Shut Down Purpose

This document will guide on, how and from whom to get approval for plant shutdown. At different situations, different approvals are defined. The bellow procedure refers the Wembiyagoda Mini Hydro Power Plant located at Kalawana, Rathnapura

Scope

The supervisor is accountable for obtaining the necessary approvals before initiating a shutdown. Obtaining prior approval is essential. In the event of an emergency, the supervisor may proceed with the shutdown, but approval must be sought through discussion and formal documentation.

Prerequisites

The estimation of downtime and generation capacity during the shutdown will be crucial and mandatory for evaluation and discussion with the DO & OM to facilitate decision- making..

Responsibilities

The responsibility for generating the permit lies with the Supervisor and the Area Engineer, while the approvals are the responsibility of the OM and DO. If the generation loss exceeds a significant threshold or extends beyond 24 hours, approval from the CEO is necessary.

Procedure

Approval Process

3. The supervisor discusses with the Area Engineer about the shut down requirement
4. The Area Engineer generates the Permit.
5. Send for OM approval.
6. Send for DO approval.
7. If the generation or number of days is considerable, get the CEO's approval.
8. Save the document in the one drive.
9. All signatures are digital. No printing

References

O & M policy

Note

21 Standard Operating Procedure – Inventory Process

Purpose

The purpose of a Standard Operating Procedure (SOP) for inventory management is to establish a set of guidelines and instructions that outline the standardized processes and best practices for managing and controlling inventory within Vidullanka Group and crucial for ensuring consistency, efficiency, accuracy, and compliance with regulations in inventory-related activities.

Scope

Procedures for receiving and inspecting incoming inventory, including verification of quantities, quality checks, and documentation of any discrepancies or damages.

Guidelines for properly storing inventory, including assigning storage locations, labeling, and ensuring appropriate handling to prevent damage or deterioration.

Processes for accurately tracking inventory movements, updating inventory records, and maintaining up-to-date information on stock levels, locations, and other relevant data as well as implementing inventory control measures, such as setting minimum and maximum stock levels, reorder points, and establishing systems for monitoring and managing stockouts, overstocking, and obsolete inventory.

Prerequisites

Review existing documentation related to inventory management, such as policies, procedures, guidelines and the system support. Assess their effectiveness, accuracy, and compliance with regulations. This review will help identify any gaps or inconsistencies that need to be addressed.

Responsibilities

Relevant plant head and AE is directly responsible for overseeing all aspects of inventory management within the organization. Responsibilities include developing inventory management strategies, setting inventory goals and targets, coordinating inventory activities, and ensuring compliance with policies and procedures.

Procedure

1.1 Receiving and Inspection Procedure:

a. Receive incoming inventory shipments.
b. Verify the items received against the purchase orders.
c. Inspect the items for quality, quantity, and compliance with specifications.
d. Document any discrepancies or damages and report them to the appropriate departments.
e. Update inventory system (PPMIS) records with the received quantities.
f. All are expected to enter the data to the system, daily from 4.30 pm to 5.pm. Reconcile monthly basis.

1.2 Storage and Organization Procedure:

g. Assign storage locations for different inventory items.
h. Label storage areas and bins for easy identification and follow the color codes guided by the Ten Second System) TSS system.
i. Follow proper handling and storage procedures for different types of inventory (e.g., heavy duty, fast movement, protections such as bearings with grease).
j. Maintain cleanliness and organization in the storage areas.

1.3 Inventory Tracking and Recording Procedure:

k. Enter new inventory items into the inventory management system.
l. Update inventory records with item details, quantities, and locations.
m. Record any inventory movements, such as transfers, returns, or disposals.
n. Conduct regular inventory counts, or cycle counts to verify physical inventory against recorded quantities.
o. Reconcile any discrepancies and adjust inventory records accordingly.

1.4 Inventory Control Procedure:

p. Determine inventory reorder levels, stock levels, and lead times. Through PPMIS system.
q. Evaluate the "Reorder level reached stock report "daily and submit for Area Engineer(AE) approval for purchasing.
r. Reorder levels are calculated and set based on the lead time and average consumption. Average consumption is calculated automatically by the system based on historical data.
s. Supervisor & AE to coordinate with purchasing officer for timely ordering and delivery.

1.5 Returns and Disposal Procedure:

a. Follow establish guidelines given by Finance for disposals.

1.6 Reporting and Analysis Procedure:

a. Generate regular reports on inventory levels, turnover, and performance metrics and discuss with Operation heads.

1.7 References

The formula we follow for calculating the unit cost is the weighted average and represented as follows:

Weighted Average = (Weighted Value1 + Weighted Value2 + ... + Weighted ValueN) / (Weight1 + Weight2 + ... + WeightN)

1.8 Note

Weighted average method is used by VLL Operations, mainly to be accordance with finance procedures as well as item prices vary over time. As new inventory purchases are made, they are added to the existing inventory, and the weighted average cost is recalculated to reflect the updated average cost per unit.

├─ Accurately forecast demand based on historical data

├─ Analyze lead time for replenishing inventory from suppliers

├─ Determine stock level to buffer against demand

├─ Calculate reorder point based on lead time and average demand

The Standard Operating Procedure (SOP) was issued by the VLL Operation Department. This department is responsible for creating and implementing guidelines, processes, and protocols to ensure smooth and efficient operations within the organization. The SOP serves as a reference document for employees to follow when carrying out their tasks and responsibilities.

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Operation Director Operation Manager